

EXPERIMENT-1

Implement and demonstrate the FIND-S algorithm to find the most specific hypothesis from the given training dataset.

Code:

```
import pandas as pd

print("="*70)
print("      FIND-S ALGORITHM")
print("="*70)

# Upload Dataset
from google.colab import files
uploaded = files.upload()

# Read Dataset
data = pd.read_csv("EnjoySport.csv")
print("\nTraining Dataset:\n")
print(data)

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
print("\nTotal Training Examples :", len(data))

# Initial Hypothesis
hypothesis = ['0'] * len(concepts[0])
print("\nInitial Hypothesis")
print(hypothesis)
```

```

step = 1
for i in range(len(concepts)):
    print("\n"+"-"*60)
    print("Training Example", step)
    print(concepts[i])

    if target[i] == "Yes":
        print("Positive Example")
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'

    else:
        print("Negative Example (Ignored in FIND-S)")
        print("Current Hypothesis :", hypothesis)
        step += 1

print("\n"+"="*70)
print("FINAL MOST SPECIFIC HYPOTHESIS")
print(hypothesis)
print("="+*70)

```

Sample Input:

```

...
=====
FIND-S ALGORITHM
=====
Choose Files EnjoySport.csv
EnjoySport.csv(text/csv) - 446 bytes, last modified: 8/5/2026 - 100% done
Saving EnjoySport.csv to EnjoySport (1).csv

```

Sample Output:

... Training Dataset:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes
4	Sunny	Cold	Normal	Weak	Warm	Same	Yes
5	Rainy	Warm	Normal	Weak	Cool	Change	No
6	Overcast	Warm	High	Strong	Warm	Same	Yes
7	Overcast	Cool	Normal	Weak	Cool	Same	Yes
8	Rainy	Cool	High	Strong	Cool	Change	No
9	Sunny	Warm	Normal	Weak	Warm	Same	Yes

Total Training Examples : 10

Initial Hypothesis

['0', '0', '0', '0', '0', '0']

Training Example 1

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Positive Example

Current Hypothesis : ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Training Example 2

['Sunny' 'Warm' 'High' 'Strong' 'Warm' 'Same']

Positive Example

Current Hypothesis : ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Training Example 3

['Rainy' 'Cold' 'High' 'Strong' 'Warm' 'Change']

Negative Example (Ignored in FIND-S)

Current Hypothesis : ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Training Example 4

['Sunny' 'Warm' 'High' 'Strong' 'Cool' 'Change']

Positive Example

Current Hypothesis : ['Sunny', 'Warm', '?', 'Strong', '?', '?']

... Training Example 5
['Sunny' 'Cold' 'Normal' 'Weak' 'Warm' 'Same']
Positive Example
Current Hypothesis : ['Sunny', '?', '?', '?', '?', '?']

Training Example 6
['Rainy' 'Warm' 'Normal' 'Weak' 'Cool' 'Change']
Negative Example (Ignored in FIND-S)
Current Hypothesis : ['Sunny', '?', '?', '?', '?', '?']

Training Example 7
['Overcast' 'Warm' 'High' 'Strong' 'Warm' 'Same']
Positive Example
Current Hypothesis : ['?', '?', '?', '?', '?', '?']

Training Example 8
['Overcast' 'Cool' 'Normal' 'Weak' 'Cool' 'Same']
Positive Example
Current Hypothesis : ['?', '?', '?', '?', '?', '?']

Training Example 9
['Rainy' 'Cool' 'High' 'Strong' 'Cool' 'Change']
Negative Example (Ignored in FIND-S)
Current Hypothesis : ['?', '?', '?', '?', '?', '?']

Training Example 10
['Sunny' 'Warm' 'Normal' 'Weak' 'Warm' 'Same']
Positive Example
Current Hypothesis : ['?', '?', '?', '?', '?', '?']

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FINAL MOST SPECIFIC HYPOTHESIS
['?', '?', '?', '?', '?', '?']

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